

MAIMOUNI YOUSSEF

Data Scientist / AI Engineer – Generative AI, NLP & RAG Systems

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Junior Data Scientist pursuing a Master's in Data Science, specialized in Generative AI, NLP, and Big Data. Strong experience in designing end-to-end AI pipelines, including data collection, preprocessing, modeling (Machine Learning, Deep Learning, Transformers, LLMs, RAG), and deployment. Experienced in building AI-powered platforms and integrating models into business applications via APIs, with workflow automation (n8n). Open to new opportunities in Data Science and AI with high impact.

PROFESSIONAL EXPERIENCE

Data Scientist / AI Engineer – TechPal

Feb 2026 – Present

- Developed an intelligent legal platform integrating automated document generation and an AI assistant based on Moroccan law and jurisprudence.
- Designed a RAG-based assistant with a hybrid pipeline (embeddings, relationships, keyword search).
- Implemented a NER model and an OCR module for extracting text from scanned legal documents, using Qdrant for semantic search.
- Modeled entity relationships using Neo4j (GraphDB) and deployed a fully local solution to ensure sensitive data privacy.

Data Scientist – SCInc, Casablanca

Jun 2025 – Sep 2025

- Designed a computer vision pipeline for retail shelf analysis.
- Implemented detection (YOLOv8), segmentation (Mask R-CNN), and classification (ResNet-50).
- Deployed solutions in a cloud environment and collaborated in an Agile team on large-scale datasets.

Développeur Web – Webagency Maroc, Casablanca

Jun 2024 – Aug 2024

- Developed full-stack applications using Laravel, React.js, and Tailwind.
- Documented REST APIs using Swagger.

PERSONAL PROJECTS

Politixpert – Moroccan Political Expert System (NLP, LLMs & RAG)

- Built an AI expert system using Moroccan political data from 6 official sources.
- Performed full NLP preprocessing: cleaning, normalization, semantic chunking.
- Generated multilingual embeddings (E5) and implemented semantic search with FAISS.
- Developed a RAG architecture with local LLM-based summarization (Qwen 1.5B).

Social Media Pipeline Automation

- Designed an automated n8n workflow for data collection, summarization, and content generation using LLMs.
- Generated social media-ready content with automated validation and deduplication.
- Built a modular architecture with error handling and human-in-the-loop validation.

SKILLS

- IA & ML** : Supervised/Unsupervised Learning, Deep Learning, Transformers, Hybrid AI Systems
- NLP & LLMs** : TF-IDF, BERT, mT5, Text Mining, Multilingual Embeddings (E5), FAISS, NER, RAG
- Computer Vision** : CNN, YOLOv8, Mask R-CNN, ResNet-50, OpenCV
- Big Data & Data Engineering** : Data collection, preprocessing, clustering (K-Means), PCA, Power BI, SSIS, Hadoop
- Automatisation & MLOps** : n8n, AI pipeline orchestration, local LLM deployment
- Développement & Cloud** : Flask, FastAPI, Laravel, ReactJS, Docker, Azure, AWS
- Bases de Données** : SQL (MySQL, PostgreSQL, Oracle), NoSQL (MongoDB), Vector DB(Qdrant), Graph DB(Neo4j)
- Langages & Outils** : Python (Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, OpenCV), Java, JS, Bash, Git
- Soft Skills** : Leadership, Problem Solving, Analytical Thinking, Teamwork, Communication
- Langues** : Arabic (Native), French (Professional), English (Professional)

EDUCATION

Master's in Data Science

2024 – 2026

- ENSA Khouribga – Sultan Moulay Slimane University

Bachelor's in Mathematics & Computer Science (Databases)

- Faculty of Sciences Aïn Chock – Hassan II University Casablanca

2021 – 2024

Awards & Certifications

- 1st Place – DevJam Hackathon (ENSIAS IT Club) – Smart Water Leak Detection (IoT + AI)
- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional - ORACLE
- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate - ORACLE
- Oracle Cloud Infrastructure 2025 Certified Foundations Associate - ORACLE
- Machine Learning with Python - IBM SkillsBuild
- Applied Data Science with Python - IBM